5 On a sunny day, the temperatures of a black tarmac road and the air above the road increase.

(a) Explain why the surface temperature of the tarmac increases.

.....

.....

..... [2]

(b) State the method of thermal energy transfer from the tarmac to the air **immediately** above the road.

..... [1]

(c) State the main method of thermal energy transfer from the air immediately above the road to the rest of the air.

..... [1]

(d) Explain why the surface temperature of the tarmac is higher than the surrounding air temperature.

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..... [2]

[Total: 6]

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